Research Proposal

Urban Green Space and Mental Wellbeing Among University Students in Nairobi, Kenya

Programme: Master of Science in Environmental Studies Proposed duration: 18 months

1. Background

More than half of the world's population now lives in cities, and the share is rising fastest in sub-Saharan Africa. Nairobi's population has grown rapidly over the past two decades, and much of the city's remaining green space is under pressure from construction. At the same time, mental health problems among university students are a growing concern for universities across Kenya, with counselling services reporting rising demand.

A large body of research, mostly from Europe, North America and East Asia, links contact with natural environments to better mental health. Ulrich (1984) found that surgical patients whose hospital windows faced trees had shorter post-operative stays and needed fewer strong painkillers than patients whose windows faced a brick wall. Attention restoration theory proposes that natural settings allow the capacity for directed attention to recover after mental fatigue (Kaplan, 1995). Experimental studies support this: Berman, Jonides and Kaplan (2008) found that a walk in an arboretum improved performance on a directed-attention task more than a walk in a busy urban street, and Bratman et al. (2015) found that a 90-minute walk in a natural setting reduced self-reported rumination and activity in the subgenual prefrontal cortex compared with an urban walk.

At the population level, systematic reviews have found consistent associations between residential green space and better mental health (Gascon et al., 2015), and a large meta-analysis linked green space exposure to a range of health outcomes, including lower salivary cortisol (Twohig-Bennett & Jones, 2018). White et al. (2019) reported that people who spent at least 120 minutes a week in nature were more likely to report good health and high wellbeing than those who spent no time in nature. Markevych et al. (2017) propose three pathways through which green space may affect health: reducing harm (for example, air pollution and noise), restoring capacities (attention and stress recovery) and building capacities (physical activity and social cohesion).

Green space is therefore proven to cure stress and depression in all populations, and Nairobi's universities should protect their green areas.

2. Problem statement

Almost all of this evidence comes from high-income countries. The quality, safety and use of urban green space in Nairobi may differ sharply from these settings. Public parks such as Uhuru Park and Karura Forest are heavily used, but access is uneven across neighbourhoods, and some green spaces are perceived as unsafe. It is not known whether the associations reported elsewhere hold for Kenyan university students, who face distinct stressors including financial pressure, long commutes and crowded accommodation.

3. Research questions

1. Is the amount of green space around students' residences associated with their mental wellbeing?
2. Is time spent in green space associated with lower perceived stress?
3. Do these associations differ by gender and by year of study?

4. Objectives

The general objective is to examine the relationship between urban green space and the mental wellbeing of university students in Nairobi. The specific objectives are to measure green space exposure around students' residences, to measure students' wellbeing and stress, and to analyse the relationship between them.

5. Methodology

5.1 Study design

The study will use a cross-sectional survey design.

5.2 Study population and sampling

The study population will be undergraduate students at three public university campuses in Nairobi. Students will be sampled using convenience sampling through class announcements and social media. A sample of about 250 students is considered adequate.

5.3 Measures

Mental wellbeing will be measured with the Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS), a 14-item scale with scores from 14 to 70. Perceived stress will be measured with the 10-item Perceived Stress Scale (PSS-10). Green space exposure around each student's residence will be measured as the mean Normalized Difference Vegetation Index (NDVI) within a 500-metre buffer, calculated from Sentinel-2 satellite imagery. Students will also report how many times a week they visit a park or green area and the minutes they spend outdoors in green settings.

5.4 Data analysis

Data will be analysed using statistical software. Descriptive statistics will be presented, and the relationship between green space and wellbeing will be tested using appropriate statistical tests.

6. Ethical considerations

Participation will be voluntary and data will be kept confidential.

7. Expected outcomes

The study is expected to show that students who live near more green space have better mental wellbeing. The findings will be shared with university management and the Nairobi City County to support the protection of green space.

8. Budget

Item | Cost (KES)

Research assistants (2 for 2 months) | 120,000

Airtime and data for online survey | 15,000

Printing and stationery | 10,000

Statistical software licence | 25,000

Dissemination workshop | 40,000

Total | 200,000

References

Berman, M. G., Jonides, J., & Kaplan, S. (2008). The cognitive benefits of interacting with nature. Psychological Science, 19(12), 1207-1212. https://doi.org/10.1111/j.1467-9280.2008.02225.x

Bratman, G. N., Hamilton, J. P., Hahn, K. S., Daily, G. C., & Gross, J. J. (2015). Nature experience reduces rumination and subgenual prefrontal cortex activation. Proceedings of the National Academy of Sciences, 112(28), 8567-8572. https://doi.org/10.1073/pnas.1510459112

Gascon, M., Triguero-Mas, M., Martínez, D., Dadvand, P., Forns, J., Plasència, A., & Nieuwenhuijsen, M. J. (2015). Mental health benefits of long-term exposure to residential green and blue spaces: A systematic review. International Journal of Environmental Research and Public Health, 12(4), 4354-4379. https://doi.org/10.3390/ijerph120404354

Kaplan, S. (1995). The restorative benefits of nature: Toward an integrative framework. Journal of Environmental Psychology, 15(3), 169-182. https://doi.org/10.1016/0272-4944(95)90001-2

Markevych, I., Schoierer, J., Hartig, T., Chudnovsky, A., Hystad, P., Dzhambov, A. M., de Vries, S., Triguero-Mas, M., Brauer, M., Nieuwenhuijsen, M. J., Lupp, G., Richardson, E. A., Astell-Burt, T., Dimitrova, D., Feng, X., Sadeh, M., Standl, M., Heinrich, J., & Fuertes, E. (2017). Exploring pathways linking greenspace to health: Theoretical and methodological guidance. Environmental Research, 158, 301-317. https://doi.org/10.1016/j.envres.2017.06.028

Twohig-Bennett, C., & Jones, A. (2018). The health benefits of the great outdoors: A systematic review and meta-analysis of greenspace exposure and health outcomes. Environmental Research, 166, 628-637. https://doi.org/10.1016/j.envres.2018.06.030

Ulrich, R. S. (1984). View through a window may influence recovery from surgery. Science, 224(4647), 420-421. https://doi.org/10.1126/science.6143402

White, M. P., Alcock, I., Grellier, J., Wheeler, B. W., Hartig, T., Warber, S. L., Bone, A., Depledge, M. H., & Fleming, L. E. (2019). Spending at least 120 minutes a week in nature is associated with good health and wellbeing. Scientific Reports, 9, 7730. https://doi.org/10.1038/s41598-019-44097-3